

2nd Laboratory Exam

FCFS Scheduling (Non-Preemptive)

User input:

```
=====
                        OPERATING SYSTEM SIMULATOR
=====
1. FCFS Scheduling (Non-Preemptive)
2. Round Robin Scheduling (Preemptive)
3. Banker's Algorithm
4. Exit
Enter choice: 1
Enter number of processes: 4
Enter arrival time for P0: 0
Enter burst time for P0: 5
Enter arrival time for P1: 1
Enter burst time for P1: 3
Enter arrival time for P2: 2
Enter burst time for P2: 8
Enter arrival time for P3: 3
Enter burst time for P3: 6
```

User Output:

```
FCFS SCHEDULING

Gantt Chart:
| P0 | P1 | P2 | P3 |
0 -> 5 -> 8 -> 16 -> 22

Process Arrival Burst  Waiting Turnaround
P0      0      5      0      5
P1      1      3      4      7
P2      2      8      6     14
P3      3      6     13     19

Average Waiting Time: 5.75
Average Turnaround Time: 11.25
```

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Round Robin Scheduling (Preemptive)

User input:

```
=====
                        OPERATING SYSTEM SIMULATOR
=====
1. FCFS Scheduling (Non-Preemptive)
2. Round Robin Scheduling (Preemptive)
3. Banker's Algorithm
4. Exit
Enter choice: 2
Enter number of processes: 4
Enter arrival time for P0: 0
Enter burst time for P0: 5
Enter arrival time for P1: 1
Enter burst time for P1: 3
Enter arrival time for P2: 2
Enter burst time for P2: 8
Enter arrival time for P3: 3
Enter burst time for P3: 6
Enter time quantum: 2
```

User Output:

```
ROUND ROBIN SCHEDULING

Gantt Chart:
| P0 | P1 | P2 | P0 | P3 | P1 | P2 | P0 | P3 | P2 | P3 | P2 |
0 -> 2 -> 4 -> 6 -> 8 -> 10 -> 11 -> 13 -> 14 -> 16 -> 18 -> 20 -> 22

Process Arrival Burst  Waiting Turnaround
P0      0       5       9       14
P1      1       3       7       10
P2      2       8      12       20
P3      3       6      11       17

Average Waiting Time: 9.75
Average Turnaround Time: 15.25
```

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Banker's Algorithm

User input:

```
=====
                        OPERATING SYSTEM SIMULATOR
=====
1. FCFS Scheduling (Non-Preemptive)
2. Round Robin Scheduling (Preemptive)
3. Banker's Algorithm
4. Exit
Enter choice: 3

BANKER'S ALGORITHM
Enter number of processes: 5
Enter number of resources: 3

Enter Allocation Matrix:
Enter 3 values separated by spaces.
P0: 0 1 0
P1: 2 0 0
P2: 3 0 2
P3: 2 1 1
P4: 0 0 2

Enter Maximum Matrix:
Enter 3 values separated by spaces.
P0: 7 5 3
P1: 3 2 2
P2: 9 0 2
P3: 2 2 2
P4: 4 3 3

Enter Available Resources (3 values): 3 3 2
```

User Output:

```
Need Matrix:
P0: [7, 4, 3]
P1: [1, 2, 2]
P2: [6, 0, 0]
P3: [0, 1, 1]
P4: [4, 3, 1]

System is in a Safe State.
Safe Sequence: P1 -> P3 -> P4 -> P0 -> P2
```

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